AP PRECALCULUS

Unit 1 • Lesson 1.12 practice • Answer key

Lesson 1.12 • Transformations of Functions

4 multiple-choice questions and 1 free-response question • Answer key with explanations

Question 1

How is the graph of $g(x) = f(x+3) - 1$ obtained from the graph of f?

(A) Right 3, down 1

(B) Left 3, down 1

(C) Left 3, up 1

(D) Right 3, up 1

Solution

Answer: (B). $x + 3 = x - (-3)$: a shift of 3 units left; -1: down 1.

Question 2

How is the graph of $g(x) = f(2x)$ obtained from the graph of f?

(A) Vertical dilation by a factor of 2

(B) Horizontal dilation by a factor of 2

(C) Horizontal dilation by a factor of $\frac{1}{2}$

(D) Vertical dilation by a factor of $\frac{1}{2}$

Solution

Answer: (C). $f(bx)$ is a horizontal dilation by a factor of $\frac{1}{b} = \frac{1}{2}$.

Question 3

The point $(4, -2)$ is on the graph of f. Which point must be on the graph of $y = 3f(x-1) + 2$?

(A) $(3, -4)$

(B) $(5, -6)$

(C) $(5, -4)$

(D) $(4, -4)$

Solution

Answer: (C). x moves right 1 to 5; y becomes $3(-2) + 2 = -4$.

Question 4

How is the graph of $y = -f(x)$ obtained from the graph of f?

(A) Reflection over the x-axis

(B) Reflection over the y-axis

(C) Reflection over the line $y = x$

(D) Shift down 1 unit

Solution

Answer: (A). Each output is multiplied by -1.

Question 5 (Free response)

Let $f(x) = x^2$ and $g(x) = -2(x-1)^2 + 8$.

(a) Describe the transformations that take the graph of f to the graph of g.

(b) Find the zeros of g.

(c) If f is restricted to the domain $[-2, 2]$ (range $[0, 4]$), find the corresponding domain and range of g.

Solution

(a) Shift right 1 unit, vertical dilation by a factor of 2, reflection over the x-axis, then shift up 8 units.

(b) $-2(x-1)^2+8=0 \iff (x-1)^2=4 \iff x=3$ or $x=-1$.

(c) Domain: $[-2+1,\, 2+1]=[-1,3]$. Range: $-2\cdot[0,4]=[-8,0]$, then $+8$: $[0,8]$.